

STAT UN 1001: Introduction to Statistical Reasoning

Syllabus | Fall 2026

Course Information

- Instructor: Tian Zheng
- TA(s): TBA
- Meeting time: Tuesday and Thursday, 10:10-11:25 a.m.
- Location: TBA
- Course platform: CourseWorks
- A [Gemini Gem for commonly asked questions on the syllabus](#) (non-archival; your questions are not visible to the instructional team.)

Course Description

This course is a friendly introduction to statistical concepts and reasoning, with emphasis on developing statistical intuition rather than mathematical rigor for its own sake. Topics include design of experiments, descriptive statistics, correlation and regression, probability, chance variability, sampling, chance models, confidence intervals, and tests of significance.

The course is designed for students who want to understand how data-based claims are made, how they can go wrong, and how statistical thinking can help in everyday life, academic work, evidence reasoning, and an increasingly AI-shaped society. Statistical reasoning is becoming a critical skill for anyone who lives, studies, and works in a world where imperfect data, algorithms, and AI systems influence information, decisions, and opportunities.

Learning Outcomes

By the end of the semester, students should be able to:

- Explain statistical ideas in plain language, using appropriate terminology.
- Distinguish good evidence from weak, misleading, or overclaimed evidence.
- Use graphs and numerical summaries to describe data accurately.
- Reason about study design, sampling, uncertainty, and chance variability.
- Interpret correlation, regression, confidence intervals, p-values, and statistical significance, for problem solving and decision making.
- Critique data-driven and AI-related claims using statistical reasoning.
- Carry out a small data project from question formulation through communication.
- Use AI tools appropriately as assistants in statistical reasoning while taking responsibility for the final work.

Course Elements

Materials

Primary textbook:

- [OpenIntro Statistics](#), available free online from OpenIntro. The course will draw on the textbook, videos, slides, datasets, labs, and selected exercises.

Project and data-science readings:

- Bin Yu and Rebecca Barter, [Veridical Data Science: The Practice of Responsible Data Analysis and Decision Making](#). We will use selected chapters and ideas, especially the data science life cycle and the PCS framework: predictability, computability, and stability.

Additional materials, datasets, videos, quiz links, and project instructions will be posted on CourseWorks.

Flipped Activities

This course uses a semi-flipped classroom format. Students will do some first-pass learning before class through short readings, videos, guided questions, and low-stakes flipped activities. Flipped activities are graded for completion, not correctness. Their main purpose is to help students prepare for class and practice engaging with statistical ideas before discussion, and to help the instructor decide what to focus class discussions on.

The normal weekly rhythm is:

- By Thursday each week: the next week's flipped activity is posted on CourseWorks.
- By Monday evening: students complete the flipped activity for completion credit.
- Tuesday class: concept discussion, Q&A, worked examples, and clarification.
- Thursday class: activities, applications, project work, or a short in-class quiz on most quiz weeks.

Lecture, Lab, And Class Activities

Class time will focus on questions, discussion, statistical reasoning activities, short demonstrations, group work, and project workshops. Statistics is learned through practice, discussion, mistakes, and revision; simply reading solutions or watching explanations is not enough.

Class participation means enhancing the quality of the learning experience for yourself and others. It includes listening carefully, respecting instructors and peers, working collaboratively, engaging with others' ideas and differences, making relevant comments, asking useful questions, and contributing to activities in good faith.

Project Sequence

The project sequence is organized around learning how to investigate data-based claims carefully. Students may explore questions of interest to them from any domain, such as health, public policy, science, education, business, sports, culture, and everyday life. AI-related examples will appear in one or two activities because they are useful for thinking about data, uncertainty, prediction, and error, but the project sequence is not an AI project sequence. We will begin with highly structured explorations of data-based claims and gradually move toward more self-directed projects.

The arc is:

- Mini-project 0 (weeks 2-3): Observing: what makes a data claim convincing?
- Mini-project 1 (weeks 3-4): Reading and Writing: *Data biography*. Where did the data come from, what is measured, and what is missing?

- Mini-project 2 (weeks 4-5): Analyzing and Visualizing: *Exploratory story*. Use visual and numerical summaries to tell a careful data story.
- Mini-project 3 (weeks 6-8): Simulating and Reasoning: *Chance and uncertainty*. Use simulation or sampling logic to assess how surprising a result is.
- Mini-project 4 (weeks 9-11): Evaluating and Critiquing: *Comparison or association*. Make and critique an inference or regression-based claim.
- Final project (due the last day of class): Putting it together: A student-chosen *data investigation* that states a question, analyzes relevant data, communicates findings, and discusses limitations.

Mini-project tasks are graded for completion. Peer reviews are graded using rubrics, since giving useful feedback is itself a course skill. Projects may receive both AI-supported feedback and human feedback, but final responsibility for the work remains with the student.

The final project is graded using a rubric. Students will have opportunities to receive feedback before the final submission, including feedback from peers, course instructional team (instructors and TAs), and approved AI-supported tools.

Assignments

Assignments include flipped activities, mini-project tasks, peer reviews, project submissions, and quiz/final-exam preparation. Quizzes and exams may draw from assigned readings, learning materials in flipped activities, class discussion, and project activities.

Quizzes

There will be six 20-minute quizzes, tentatively scheduled for:

- Quiz 1: Thursday, October 1
- Quiz 2: Thursday, October 15
- Quiz 3: Thursday, October 29
- Quiz 4: Thursday, November 12
- Quiz 5: **Tuesday**, November 24
- Quiz 6: Thursday, December 3

Quizzes will emphasize interpretation, reasoning, and core computations. They will be paper-based and will use question formats similar to the cumulative final exam, so that students get repeated practice with the kinds of reasoning they will need later. Quiz questions will usually be more focused and less variable in difficulty than final exam questions. The lowest quiz score will be dropped.

Final Exam

The final exam will be cumulative and scheduled according to the university final exam schedule. It will emphasize statistical reasoning, interpretation, and the ability to connect concepts across the semester. The final exam will be paper-based and similar in format to the quizzes, but with more variability in difficulty, including some questions that require synthesis across topics.

Computation

This course emphasizes statistical intuition. Computation will be used to make ideas visible.

Students will be guided to use accessible tools for data work, such as Tableau, interactive dashboards, Google Sheets, guided Jupyter notebooks/Google Colab, and Posit Cloud. The exact tool stack will be introduced gradually during the activities segment of the lectures. Students will receive instructions, templates, and starter code/notebooks where appropriate.

Evaluation

Grading breakdown:

- Flipped activities: 10%
- Six biweekly in-class quizzes, lowest score dropped: 20%
- Mini-project tasks and peer review: 10%
- Final project: 20%
- Cumulative final exam: 40%

There is no midterm exam. The biweekly quizzes are short, focused assessments intended to keep learning steady and to make the final less mysterious.

Weekly Schedule

The following weekly plan is tentative as of *June 14th, 2026* and may change as the course develops.

Week	Dates	Statistical Focus	Project / Application Focus	Main Resources
1	Sep 8, Sep 10	What is statistics? Data, variables, cases, claims, and context	Course onboarding; data claims in everyday life	OpenIntro Ch. 1; VDS Ch. 1
2	Sep 15, Sep 17	Study design, sampling, bias, confounding, and measurement	Mini-project 0; data biography launch	OpenIntro Ch. 1; VDS Ch. 2
3	Sep 22, Sep 24	Describing and visualizing numerical and categorical data	Mini-project 0 workshop; no major grade-bearing assessment	OpenIntro Ch. 2
4	Sep 29, Oct 1	Comparison, variability, probability foundations, and Quiz 1	Mini-project 1 due; Mini-project 2 launch; Quiz 1 on Thursday	OpenIntro Ch. 2-3; selected VDS Ch. 4-5
5	Oct 6, Oct 8	Conditional probability, risk, simulation, and chance models	Mini-project 2 due; base rates, false positives, and risk claims	OpenIntro Ch. 3
6	Oct 13, Oct 15	Random variables, normal/binomial models, sampling distributions, and Quiz 2	Mini-project 3 launch; Quiz 2 on Thursday	OpenIntro Ch. 3-4

Week	Dates	Statistical Focus	Project / Application Focus	Main Resources
7	Oct 20, Oct 22	Confidence intervals and sampling variability	Mini-project 3 workshop; uncertainty in claims	OpenIntro Ch. 5
8	Oct 27, Oct 29	Confidence intervals, hypothesis testing, and Quiz 3	Mini-project 3 due; Quiz 3 on Thursday	OpenIntro Ch. 5
9	Nov 3 (no class), Nov 5	Hypothesis testing and statistical significance	Mini-project 4 launch; claims about performance, improvement, and evidence	OpenIntro Ch. 5
10	Nov 10, Nov 12	Inference for proportions and categorical data	Quiz 4 on Thursday; peer review workshop	OpenIntro Ch. 6
11	Nov 17, Nov 19	Inference for numerical data and regression preview	Mini-project 4 due; final project proposal clinic	OpenIntro Ch. 7-8
12	Nov 24, Nov 26 (no class)	Correlation, regression, and Thanksgiving week	Quiz 5 on Tuesday; prediction, error, and generalization	OpenIntro Ch. 8
13	Dec 1, Dec 3	Regression, responsible communication, and review	Final project workshop; Quiz 6	OpenIntro Ch. 8-9; VDS wrap-up

AI Use

AI will appear in three related ways. First, we will study AI-related claims and systems as one important subject of statistical inquiry: What data are used? What errors are possible? What uncertainty is reported or missing? What decisions might follow? Second, students may use AI tools as assistants for brainstorming, feedback, coding help, revision, and explanation, while remaining responsible for their own reasoning and final work. Third, the instructional team may use AI tools for operational support, such as drafting examples, organizing feedback, or identifying common themes in student questions. When AI is used in ways that affect student-facing materials or feedback, we will make that use transparent.

The course treats AI neither as magic nor as something to ban or ignore. Statistics is one of the pillars of AI development and also offers important ways to understand and use AI rigorously and responsibly. We will use statistical reasoning to ask what AI-related claims can and cannot show, and we will practice using AI tools transparently and responsibly when they can support learning or help the course run more efficiently.

AI Policy

AI tools may be used in this course for brainstorming, feedback, coding help, revision, and generating alternative explanations. They may not replace the student's own reasoning, interpretation, or responsibility as a learner.

Students must acknowledge meaningful AI use on project work. A suitable acknowledgment includes:

- What tool was used.
- What it was used for.
- What the student changed, checked, rejected, or verified.

The student is responsible for all parts of the final submission.

Example: *"I used ChatGPT to brainstorm possible visualizations and to debug an R error. I checked the code output against the dataset, revised the plot labels to make it easier to read, and wrote the interpretation without AI-generated text."*

AI-generated claims, code, and citations can be wrong. Students are responsible for checking them.

Undisclosed meaningful use of AI on submitted work is a violation of academic integrity and will be reported if detected. Students should ask the instructional team before submitting work if they are unsure whether a particular use of AI needs to be acknowledged.

Other Policies

CourseWorks, Email, And Questions

Students are responsible for staying informed about course expectations, deadlines, and announcements. Please check CourseWorks at least once each day during the semester. Important time-sensitive announcements will also be sent by email, so students are expected to monitor their Columbia email regularly.

Students are encouraged to ask questions openly on the CourseWorks discussion board when the question could be useful to other students. This includes questions about course logistics, concepts, readings, flipped activities, projects, quizzes, and statistical software. Posting questions publicly helps the instructional team answer once in a place where everyone can benefit from the response. Direct email should be reserved for matters that require a private conversation, such as personal circumstances, accommodations, illness, grading concerns, or other sensitive issues.

Academic Integrity

Academic honesty is expected of all students at all times. Students are responsible for understanding and abiding by the academic integrity standards of their school, including Columbia College and SEAS's [Academic Integrity](#) statements, General Studies' [Honor Pledge](#), and Barnard's [Honor System](#), as applicable. Ignorance of these policies is not a defense in academic integrity proceedings.

All submitted work must be the student's own work, or the work of the student's approved project group when group work is explicitly assigned. Students must properly acknowledge the ideas, sources, tools, and assistance that shaped their work. Cheating, plagiarism, unauthorized collaboration, fabrication of data or sources, and misrepresentation of another person's or tool's work as one's own are violations of academic integrity.

Late Registration And Catch-Up

Students who register late should contact the course staff as soon as possible. The first three weeks are designed to make late entry manageable. Early preparation activities may be completed late for completion credit during the catch-up window announced on CourseWorks.

Accessibility And Belonging

Statistics is learned through practice, discussion, mistakes, and revision. Students are expected to contribute to a respectful learning environment where questions are welcome and uncertainty is treated as part of the work.

Students who need accommodations should follow university procedures and contact the instructor early so arrangements can be made.